



TREEMAP

ALSO KNOWN AS Heat map (wrongly)



REPRESENTATION DESCRIPTION

A treemap is an enclosure diagram providing a hierarchical display that shows how quantitative values for different constituent categorical parts make up a whole. It uses a contained rectangular layout, often termed **squarified**, representing the 100% total. This is divided into proportionally sized shape marks (rectangular tiles) for the quantitative values associated with each categorical part. The organisation of each shape is based on a tiling algorithm to optimise the overall space usage and to cluster related categories into larger rectangle-grouped containers. Attributes of colour are often used to represent further quantitative measures or categorical associations. Treemaps are most commonly used, and of most value, when there are many parts to the whole. The constituent parts must be exclusive and the total representative of a meaningful 'whole', otherwise the chart will be corrupted.

EXAMPLE Comparing the relative value and daily changes of market capital for stocks across the S&P 500 index grouped by sectors and industries.



Figure 6.26 Finviz: Standard and Poor's 500 Index Stocks (www.finviz.com)

PRESENTATION TIPS

INTERACTIVITY: Interactive features that enable selection events to trigger annotated tooltips can be useful, providing direct value labels and details. There may also be scope for modifications to temporal dimensions, changing the sizes and colouring accordingly, or zooming techniques to get a closer view of small constituent parts.

ANNOTATION: Group or container labels can be hard to allocate space to efficiently, so borders are usually applied to indicate the relevant enclosure areas. Effective direct value labelling becomes difficult as the rectangles get smaller, so only the most prominent values may be annotated, especially if interactivity is not available. Any colours used must be explained through the inclusion of a legend.

COMPOSITION: As the tiling algorithm used by any given tool to create a treemap will be focused on optimising the dimensions and arrangement of the rectangular shapes, treemaps may not always be able to facilitate meaningful sorting of high to low values within each enclosure. However, you will generally find larger areas appear in the top left and work outwards towards the smaller constituent parts.

VARIATIONS & ALTERNATIVES

A variation of the treemap sees the overall rectangular layout replaced by a circular one and the tiles represented by polygonal shapes. These are known as 'Voronoi treemaps' as the tiling algorithm is informed by a Voronoi tessellation. The 'circle packing diagram', a variation of the 'bubble chart', similarly shows many parts to a whole but uses non-tessellating circular shapes. The 'Marimekko chart' is similar in appearance to a treemap but, in contrast to the treemap's hierarchical display, presents a breakdown of quantitative percentages and/or absolute values across two categorical dimensions.



NETWORK DIAGRAM

ALSO KNOWN AS Node-link diagram, graph, hairball graph, social network

C H **R** T S

CONNECTIONS

REPRESENTATION DESCRIPTION

Node-link diagrams display relationships through the connections between data items. The common version of this type of diagram displays items as nodes, represented by point marks, with links or edges (represented by lines) depicting the existence of a connection. These connecting lines will sometimes encode an attribute of direction to indicate the influencer relationship. Replacing point marks with a geometric shape and using attributes of size is a further variation. In some versions a further quantitative weighting is applied to show the relationship strength, maybe through increased line width. Attributes of colour may be used to indicate a quantitative value (e.g. number of connections) and/or some categorical classification.

EXAMPLE Exploring the connections of voting patterns for Democrats and Republicans across all members of the US House of Representatives from 1949 to 2012.

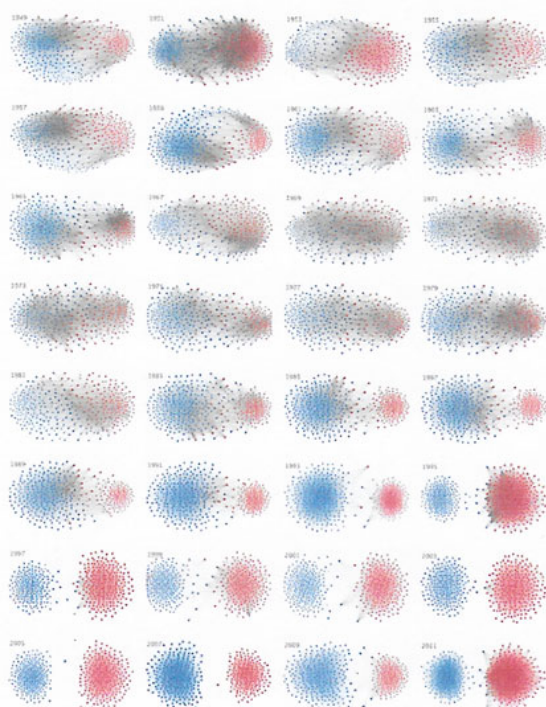


Figure 6.32 The Rise of Partisanship and Super-cooperators in the U.S. House of Representatives, visualisation by Mauro Martino, authored by Clio Andris, David Lee, Marcus J. Hamilton, Mauro Martino, Christian E. Gunning, and John Armistead Selde

VARIATIONS & ALTERNATIVES

There are many derivatives of the node-link diagram, as explained, based on variations in the use of different attributes. 'Hive plots' and 'BioFabric' offer alternative approaches based on replacing nodes with vertices.

PRESENTATION TIPS

INTERACTIVITY: Node-link diagrams are particularly useful when offered with interactive features, enabling the user to interrogate and manipulate the display to facilitate visual exploration. The option to apply filters to reduce the busyness of the display, and maybe enable isolation of individual node connections, can help viewers to focus on specific parts of the network rather than face the whole system at once.

ANNOTATION: The complexity revealed by these diagrams is often a reflection of the underlying subject complexity, so it can be helpful to use annotation to surface key observations about significant clusters or label those items with the most connections.

COLOUR: Aside from the possible categorical colouring of each node, decisions need to be made about the colour of the connecting lines, especially with regard to how prominent these links will be in contrast to the nodes.

COMPOSITION: Composition decisions will be so varied for any network diagram depending on the complexity and volume of the data and the output constraints around space and consumption format. There are several common algorithmic treatments used to compute custom arrangements to optimise network displays, such as force-directed layouts using the physics of repulsion and springs to amplify relationships. There are also simplifying techniques such as edge bundling to aggregate or summarise multiple similar links.



CHORD DIAGRAM

ALSO KNOWN AS Radial Sankey diagram, radial network, arc diagram

C H R T S

CONNECTIONS

REPRESENTATION DESCRIPTION

A chord diagram displays relationships through connections between and within category items. The diagram is formed around a radial display with categories located around the edge: either shown as individual nodes or as arc segments proportionally sized around the circumference to represent a part-to-whole breakdown. Emerging inwards from each origin position are curved lines that join with related categorical destinations around the edge. The connecting lines are proportionally sized according to a quantitative measure and a directional or influencing relationship is often indicated. Attributes of colour hue are commonly used to distinguish different category groupings visually.

EXAMPLE Exploring the connections of migration between and within ten world regions based on estimates across five-year intervals between 1990 and 2010.



Figure 6.34 The Global Flow of People, by Nikola Sander, Guy J. Abel and Ramon Bauer

PRESENTATION TIPS

INTERACTIVITY: Chord diagrams are particularly useful when offered with interactive features, enabling the user to interrogate and manipulate the display to facilitate visual exploration. The option to apply filters to reduce the chaos of the visual and enable isolation of individual or groups of flows helps users to focus on specific relationships of interest. Interactively enabled labelling can also be beneficial as direct labelling is difficult to incorporate elegantly.

ANNOTATION: Direct labelling will normally be restricted to noteworthy points only. Any colours used must be explained through the inclusion of a legend.

COLOUR: Aside from the categorical colouring of each node, decisions need to be made about the colour of the connecting lines, especially on deciding how prominent these links will be in contrast to the nodes. Sometimes the connections will match the origin or destination colours, or they will combine the two (with a start and end colour blend to match the relationship).

COMPOSITION: The main arrangement decisions concern sorting. Firstly, by deriving as much logical meaning from the categorical values within each stack and, secondly, by deciding on the sorting of the connecting lines in the z-dimension – if many lines are crossing, there is a need to think about which will be on top and which will be below. Showing the direction of connections can be achieved using arrowheads or colour lightness variation. One common, subtle solution is to pull the destination join away from the edge of the destination arc to contrast with connecting lines that emerge directly from an origin.

VARIATIONS & ALTERNATIVES

Variations of the chord diagram would be to consider using a single baseline axis, placing all category items along it and forming connections between using semi-circular arcs. Additionally, a 'Sankey diagram' would be relevant if there are distinct origins and destination relationships to reveal.